

BLOG – MICROSOFT COPILOT

Introduction

What is Microsoft Copilot?

Microsoft Copilot is an AI-powered assistant developed by Microsoft that helps users perform tasks, generate content, answer questions, write code, and improve productivity. It combines Large Language Models (LLMs) with Microsoft's services and applications.

Key Features

- ✓ Content Generation
 - ✓ Code Assistance
 - ✓ Document Summarization
 - ✓ Productivity Enhancement
 - ✓ Context-Aware Responses
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How Copilot Works?

Microsoft Copilot is built using advanced Large Language Models and integrates with Microsoft services such as Word, Excel, PowerPoint, Outlook, and GitHub.

Core Process

✓ User Input

- User enters a prompt or question.

✓ Context Collection

- Copilot gathers relevant information from documents, emails, files, or code repositories.

✓ AI Processing

- The LLM analyzes the prompt and context to understand user intent.

✓ Response Generation

- Copilot generates text, code, summaries, or recommendations.

✓ Output Delivery

- The final response is displayed to the user.
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Architecture

Input Layer

- User Prompt
- Context Data
- Application Data

Processing Layer

Tokenization

Input text is divided into tokens.

Example:

"I love AI" → ["I", "love", "AI"]

Embedding

Tokens are converted into numerical vectors.

Positional Encoding

Position information is added to maintain word order.

Transformer Layers

- Self-Attention
- Feed Forward Network
- Layer Normalization
- Residual Connections

Output Layer

- Probability Calculation (Softmax)

- Response Generation
- Final User Output

Background of Microsoft Copilot

Microsoft introduced Copilot to bring AI assistance directly into productivity tools and development environments. It was developed through Microsoft's partnership with OpenAI and integrates advanced AI models into everyday workflows.

Key milestones:

- GitHub Copilot launched in 2021.
- Microsoft 365 Copilot introduced for Office applications.
- Copilot expanded across Windows, Edge, and Microsoft services.

Large Language Model (LLM)

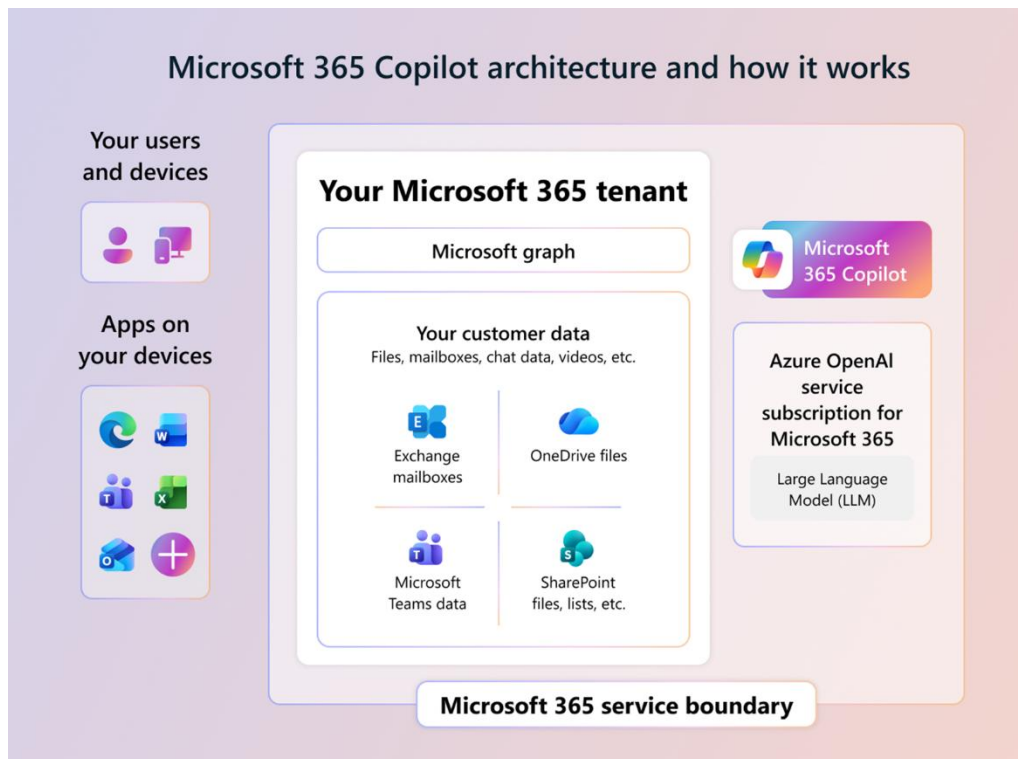
A Large Language Model (LLM) is an AI model trained on massive datasets that can understand and generate human-like text.

Benefits of LLMs

- Natural Language Understanding
- Content Creation
- Code Generation
- Translation
- Question Answering

Examples:

- ChatGPT
- Microsoft Copilot
- Claude Google Gemini



Official Microsoft Documentation

- [Microsoft 365 Copilot Architecture Overview](#)
- [Microsoft 365 Copilot Overview and Processing Flow](#)

Processing of Microsoft Copilot

1. User Prompt

The process begins when a user enters a query or instruction.

Example:

"Summarize my project report and create a presentation."

2. Prompt Understanding

Copilot analyzes the prompt to understand the user's intent, context, and required task. It identifies important keywords and determines what information is needed.

3. Grounding

Grounding means collecting relevant context before generating a response. This helps the AI produce accurate and personalized results.

4. Microsoft Graph Retrieval

Copilot accesses Microsoft Graph to retrieve relevant information from:

- Emails
- Documents
- Calendar events
- Teams chats
- OneDrive files

This ensures that the response is based on the user's available data.

5. Prompt Augmentation

The retrieved information is combined with the original user prompt. This enriched prompt provides additional context to the AI model.

Example:

Original Prompt:

"Create a summary of my project."

Enhanced Prompt:

"Create a summary of my project using Project_Report.docx and meeting notes from Teams."

6. Large Language Model Processing

The enhanced prompt is sent to the Large Language Model (GPT-based model). The model processes the information using Transformer architecture and generates an appropriate response.

Key operations include:

- Tokenization
- Embedding
- Self-Attention Mechanism
- Context Understanding
- Response Generation

7. Safety and Compliance Checks

Before displaying the response, Microsoft applies security, privacy, and compliance checks to ensure that the generated content is safe and follows organizational policies.

8. Response Generation

The final output is generated and formatted according to the requested task.

Examples:

- Summary
- Email Draft
- PowerPoint Content
- Code Suggestions
- Data Analysis

9. User Output

The generated response is displayed to the user through Microsoft Copilot.

Complete Flow

User Prompt



Prompt Understanding



Grounding



Microsoft Graph



Prompt Augmentation



Large Language Model (GPT)



Safety & Compliance Check



Response Generation



User Output